

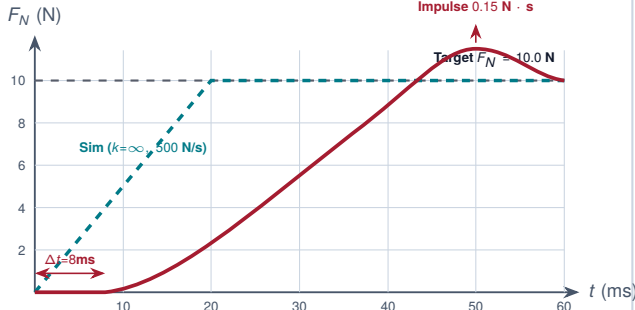
THE MULTIDIMENSIONAL SIM-TO-REAL REALITY GAP VECTOR $\Delta = [\Delta_{\text{dyn}}, \Delta_{\text{lat}}, \Delta_{\text{sens}}, \Delta_{\text{vis}}]^T$

Task-relevant mismatches across heterogeneous physical channels: each carries distinct units, failure modes, runtime detectors, and unclosable residuals.

CHANNEL 1: DYNAMICS & CONTACT (Δ_{dyn})

Units: N, N/s, rad/(N · m)

Overshoot 11.5 N
Impulse 0.15 N · s



Physical Failure Signature: Delayed force rise interpreted as non-contact → commands extra motor torque → 3.0 mm over-travel into current trip (l_{max}).

Runtime Detector: Torque/force derivative clamp ($dF/dt \leq 2.5 \times 10^4$ N/s) in MCU reflex loop.

Unclosable Residual: Gearbox compliance (1.5×10^{-3} rad/N · m) & contact damping.

CHANNEL 2: ACTUATOR & LATENCY (Δ_{lat})

Units: ms, mm

Sim Assumption: Synchronous Cycle ($\Delta t = 20$ ms, Latency = 0 ms)

Sim Step k (Instant Actuation)

Physical Hardware Reality: Segmented Pipeline Latency



Blind Travel: $\Delta x = v \cdot \Delta t = 0.20$ m/s $\times$ 29.5 ms = 5.9 mm unbraked displacement

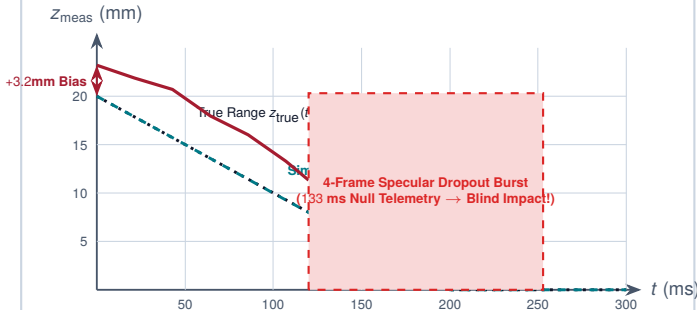
Physical Failure Signature: Commands arrive 29.5 ms late → 5.9 mm displacement depletes phase margin → high-frequency limit-cycle oscillation and impact shocks.

Runtime Detector: Hardware timestamp age watchdog and RTOS jitter monitor.

Unclosable Residual: Inference contention tail ($P_{99} = 49.5$ ms, jitter ± 6.2 ms).

CHANNEL 3: SENSING & TRANSDUCTION (Δ_{sens})

Units: mm, N, lux



Physical Failure Signature: +3.2 mm bias triggers premature regulation 3.2 mm above surface; 133 ms optical dropout blinds policy during descent → unbraked crash.

Runtime Detector: Consecutive dropout frame counter and thermal drift estimator.

Unclosable Residual: Thermal drift (0.40 N/ 10° C), noise floor ($\sigma = 1.8$ mm).

CHANNEL 4: PERCEPTUAL / VISUAL (Δ_{vis})

Units: Cosine Dist, mm

Pixel-Space Metric (Misleading)

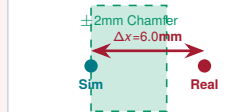
Mean Intensity $\Delta I = 5.5\%$ (MSE = 0.002)



$\Delta_{\text{pixel}} \leq 5.5\%$ (Appears Converged)

Latent Spatial Activation (True Gap)

ViT Patch Token Centroid Shift:



Exceeds 2.0mm Chamfer → Collision

Physical Failure Signature: 5.5% pixel loss masks 6.0 mm spatial token offset → end-effector strikes fixture shoulder instead of bore.

Runtime Detector: Feature-space Mahalanobis distance monitor on encoder tokens.

Unclosable Residual: Specular BRDF reflection and micro-texture variations.

Systems Architectural Synthesis:

The sim-to-real gap is inherently a 4D coupled vector $\Delta = [\Delta_{\text{dyn}}, \Delta_{\text{lat}}, \Delta_{\text{sens}}, \Delta_{\text{vis}}]^T$. Sizing the gap requires paired-trace telemetry across each physical axis. Domain randomization broadens coverage over modeled parameters (Ξ) but cannot bridge structural omissions or unmodeled timing tails outside the simulation graph.